
Dissecting Higgs TTS v3: Layer-Resolved Linear Readability of Speaker Attributes in a Voice-Cloning Audio LLM

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Abstract

We ask where a voice-cloning audio language model represents the attributes of the voice it is cloning. During generation by a 4B-parameter, 36-layer autoregressive TTS model (Higgs-Audio v3), we capture one mean residual-stream vector per layer per clip with read-only hooks, and slide held-out linear probes up the stack for five attribute views (languages, gender, accents, and emotion heard from a reference recording vs. told via the model’s native control token). Attributes become linearly readable in a consistent depth order — told emotion and gender near-ceiling mid-stack, language 98.9% at layer 19 — while emotion heard from an emotional recording averages only 61.7% at its best layer, completing a hear/tell 2×2 (chance 20% / 61.7% / 98.3% / 99.5%) that survives actor-disjoint evaluation. With fully independent audio, told-only and voice-only directions converge (raw cos up to 0.90) — but calibration we argue should be standard for such claims (split-half reliability ceilings, cross-emotion specificity, anisotropy-matched nulls, sex-composition controls) shows the shared latent is coarse: off-diagonal alignment reaches 0.93, the matched null is 0.23, and the previously headlined “sad exception” (0.445) dissolves under reliability correction. Language and gender directions are near-orthogonal at every layer, consistent with compositional cloning; a language probe transfers across corpora; and a leave-one-speaker-out test converts our flashiest number — a 100% accent probe — into a measured diagnostic (collapse to 67.7%, restoring genuine mid-stack structure). Acoustic-feature baselines sepa-

rate what the model computes from what the stimulus already contains. All measurements are observational — read-only maps of what layers encode, not claims about causal use.

1. Introduction

A voice-cloning text-to-speech (TTS) model receives a few seconds of reference audio and a transcript, and must implicitly answer a stack of questions before emitting its first audio token: what language is this? who is speaking? are they male or female? how do they feel? Modern voice cloners answer these questions well enough to speak almost any language in almost any voice, but where in the network those answers live — and whether attributes specified through different interfaces share representations — has received far less attention than the analogous questions for text LLMs (Alain & Bengio, 2017; Belinkov, 2022) and self-supervised speech encoders (Pasad et al., 2021).

We map the residual stream of Higgs-Audio v3, a 4B-parameter autoregressive audio LLM with a 36-layer Qwen3-style backbone (Qwen Team, 2025), while it performs real voice-cloning synthesis. The instrument is deliberately minimal: one mean residual vector per layer per generated clip, read-only forward hooks, then held-out linear probes and discriminant geometry on top. This yields a layer-resolved map of five attribute views — language, gender, accent, and emotion with and without the model’s own control token — that we summarize as an identity ladder (Figure 1).

Our contributions:

- A depth-ordered readability map, cross-validated (§4.1). Speaker attributes become linearly decodable at characteristic, ordered depths — token-conditioned emotion $\approx 95\%$ by layer 11 and at ceiling mid-stack, gender and language mid-stack — with 95% bands over 100–300 repeated splits and label-shuffled permutation nulls; emotion inferred from an emotional recording averages 61.7% [47–77] at its best layer and never saturates.

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- A hear/tell 2×2 for emotion, and a calibrated convergence result (§4.2). Hearing alone: 61.7%. Hearing + token (audio held constant): 98.3%. Token on a neutral carrier (told-only, corrected capture): 99.5%, and 100% under speaker-disjoint CV — emotional audio adds nothing to a clean instruction. With fully independent audio the two routes’ directions converge (raw cos 0.76–0.90 over five emotions at layer 6) — but calibration shows the convergence is coarse: the cross-emotion off-diagonal is nearly as high (0.68 vs. 0.77), an anisotropy-matched null sits at 0.23, part of the raw value is a shared sex-composition component, and the previously headlined “sad exception” (0.445) dissolves under split-half reliability correction — sad’s direction is weak, not misaligned.
- Geometry consistent with compositionality, at every layer (§4.3). Language and gender directions are near-orthogonal across all 36 layers (mean $|\cos| \in [0.113, 0.215]$, raw vectors; norm retention $\geq 97.0\%$ after projecting gender out). Mid-stack (layers 8–28) the worst pair of the six language directions peaks at $|\cos| \approx 0.51$, before an ≈ 13 – $19\times$ norm explosion (layer 34 vs. layer 12) reorganizes the geometry in the deepest layers.
- A measured diagnostic for probe artifacts (§4.4). Our flashiest number — a 100% accent probe from layer 11 onward, flagged by its flat layer profile and 3-speakers-per-accent provenance — is now tested: under leave-one-speaker-out evaluation it collapses to 67.7% (chance 25%) at layer 20, and the collapse restores the rising-then-peaking layer structure of a genuine content attribute. Speaker-disjoint evaluation is standard in paralinguistics (Schuller et al., 2009); we re-issue it for layer-resolved probing, paired with a shape diagnostic (a flat layer profile as the tell) specific to that setting.

Everything here is a map of what layers encode, not a claim about what the model causally uses; intervention-based analysis of the same model is reported elsewhere and deliberately excluded here.

2. Related Work

Probing. Linear probes are the standard instrument for asking what a representation makes linearly explicit (Alain & Bengio, 2017), with well-known caveats along three axes: probe capacity and dataset artifacts (Belinkov, 2022; Hewitt & Liang, 2019), information-vs-extractability trade-offs (Voita

& Titov, 2020; Pimentel et al., 2020), and the gap between what a layer encodes and what the model uses (Elazar et al., 2021). Our deliberately low-capacity PCA(64)+ridge probe with label-shuffled permutation nulls follows the spirit of this literature: the null bounds what the fixed pipeline extracts from structureless labels, complementary to control-task selectivity (Hewitt & Liang, 2019).

Layer-wise analysis of speech models. A substantial encoder tradition maps where speech attributes live by depth: layer-wise probing of wav2vec-style models (Pasad et al., 2021; 2023), SUPERB layer-weight analyses (Yang et al., 2021), WavLM’s speaker-vs-content layer findings (Chen et al., 2022), language identity in SSL representations (de Seyssel et al., 2022), and layer-resolved speech emotion recognition (Wagner et al., 2023). The speaker-early/content-mid ordering we observe in a generative decoder echoes what those studies report for SSL encoders; what is new here is (i) capture during autoregressive voice-cloning synthesis in a codec-token TTS model — to our knowledge the first layer-resolved probing of a production voice cloner during generation — (ii) the token-vs-voice conditioning routes, and (iii) the depth-resolved direction geometry.

Linear representations and cross-route convergence. The view that high-level features occupy linear directions in the residual stream underlies much recent interpretability work (Park et al., 2024; Zou et al., 2023), including cross-model convergence (Huh et al., 2024). The closest analogues of our two-interface result are multimodal neurons (Goh et al., 2021) — concept-level convergence across input modalities — and function/task vectors (Hendel et al., 2023; Todd et al., 2024), where instructions and demonstrations induce coinciding representations in text LLMs. Our result is an audio-generation analog: a control token and an acoustic exemplar writing overlapping residual directions, measured layer-by-layer with the calibration (reliability ceilings, cross-emotion specificity, matched nulls) that Section 4.2 shows is essential before reading such cosines at face value. The projection-out tool we use for cross-view geometry has its own pedigree in the gender-direction and nullspace-projection literature (Bolukbasi et al., 2016; Ravfogel et al., 2020); multilingual direction geometry has likewise been mapped in text models (Chang et al., 2022).

Expressive TTS and attribute disentanglement. A separate engineering line builds the two interfaces we measure: style tokens and prosody transfer put “tell” and “hear” controls into TTS by construction (Wang et al., 2018; Skerry-Ryan et al., 2018), AutoVC disentangles

speaker from content through bottlenecks (Qian et al., 2019), and prompt-controlled TTS makes the “tell” interface natural language (Guo et al., 2023; Yang et al., 2024). Those works engineer separate interfaces or disentangled factors at training time; this paper measures, layer by layer, whether a production model’s two native interfaces (control token vs. reference audio) converge internally — and finds convergence that is real but coarse (§4.2).

Audio LLMs. Codec-token language models now dominate zero-shot TTS (Wang et al., 2023). Higgs-Audio v3 (Boson AI, 2026) conditions on a reference voice and transcript and generates interleaved audio tokens; its backbone exposes a standard residual stream, making text-LLM interpretability tooling directly applicable. Probing of audio models has concentrated on SSL encoders and, recently, audio-understanding LLMs; the generative, voice-cloning side is nearly untouched.

3. Setup

Model and capture. We run bosonai/higgs-audio-v3-tts-4b (36-layer Qwen3-style backbone (Qwen Team, 2025), hidden size 2560) via its community transformers port, in bf16 on a single Apple M1 Max (MPS). During each voice-cloning generation we attach read-only forward hooks to all 36 decoder layers and store the sequence-mean residual-stream vector per layer, giving one 36×2560 summary per clip. Hooks are removed after capture; no weights or activations are modified anywhere in this study.

Attribute views and data. Five views over four corpora: (i) language: six FLEURS languages (Conneau et al., 2022) (en, de, fr, ko, ja, zh), residuals captured during monolingual production synthesis (reused from a prior capture of the same model); (ii) gender: male vs. female FLEURS reference voices (English carrier for both groups), labels taken from the FLEURS meta-data gender column; (iii) accent: four English accents (US, British, Indian, Irish) from a VCTK mirror (Yamagishi et al., 2019) (mic1; 3 speakers per accent); (iv–v) emotion, two conditions: five RAVDESS emotions (Livingstone & Russo, 2018) (neutral, happy, angry, sad, surprise), probed as the emotional reference recording alone, or as the same recording with the model’s native single-token control `<|emotion:·|>` prepended and a neutral reference transcript (20 clips per emotion per condition; the same 20 recordings underlie both) — an audio-held-constant $\pm$ token contrast, the result of a capture-code discovery narrated in §4.2. A corrected re-capture adds the told-only condition (token on a speaker-matched neutral carrier), and a second revision extends the emotion set with

fearful and disgust (both single native tokens) in both conditions.

Notation. For clip i of a view, $x_i(\ell) \in \mathbb{R}^{2560}$ is the sequence-mean residual at layer $\ell \in \{0, \dots, 35\}$. For class c under conditioning route r (“voice” = clone an emotional recording; “tag” = the model’s control token), $\mu_c^r(\ell)$ is the class mean and $v_c^r(\ell) = \mu_c^r(\ell) - \mu_{\text{neutral}}^r(\ell)$ the class’s shift vector (one-vs-rest diff-of-means for the non-emotion views).

Lenses. Per layer ℓ and view, on the mean residuals: (a) probe accuracy: a ridge classifier (one-hot least squares, $\lambda=1$) on PCA features — $k = \min(64, n_{\text{train}}-1)$, so gender’s effective k is 27–33 — trained on a derive split and evaluated held-out (language $n_{\text{test}}=120$; emotion $n=30$ per route; accent $n=24$; gender $n=8$); Section 4.5 shows the results are insensitive to k , λ , and probe family; (b) Fisher separation: between-class over within-class scatter along discriminant directions; (c) discriminant η^2 : in-sample separated-variance fraction after a regularized LDA in a 48-dim PCA subspace (used only for visualization — we read its shape across layers, not its absolute value); (d) route convergence (emotion only): $\cos(v_E^{\text{tag}}(\ell), v_E^{\text{voice}}(\ell))$, reported with three calibrations the revision adds — split-half reliability of each side (the noise ceiling), the cross-emotion off-diagonal $\cos(v_{E'}^{\text{tag}}, v_E^{\text{voice}})$ (the specificity control), and an anisotropy-matched null (random-subset diff-of-means pairs); (e) cross-view orthogonality: mean $|\cos|$ between the six language one-vs-rest mean-offset directions and the gender direction, plus the norm fraction each retains after projecting out the other’s subspace (all 36 layers in the revision).

Data details. All selection is deterministic — sorted file lists, duration filters, and seed-0 shuffles; splits are disjoint by construction and nothing was hand-picked. Language: 100 FLEURS train clips per language (duration 4–20s, first 50 per gender in archive order — exactly 50M/50F per language, verified against the corpus manifest), captured during self-clone synthesis (each clip’s own voice and transcript; max 256 new tokens, first 8 generated tokens dropped); probe split 40/20/8. FLEURS ships no speaker IDs, so clips are treated as pseudo-speakers. Gender: 20 clips per group from the en_us pool (English carrier for both groups; 14 captured originally, 6 added in the revision top-up), every voice speaking from the same 8 shared target sentences (max 200 tokens); the one-shot maps used derive 10 / test 4 on the first 14. Emotion: RAVDESS speech set (24 actors, 12M/12F, North-American English, two fixed sentences); a seed-0 draw of 20 clips

Table 1. The ladder, quantified. CV peak = mean over repeated stratified splits (95% percentile band = split variability, not a CI); 5-fold CI = Clopper–Pearson 95% interval from pooled 5-fold predictions at the peak layer (each clip tested once); Acou. = the model-free acoustic-feature baseline on the same inputs (§4.5); n = clips per view.

View	CV peak [band]	L	5-fold CI	Ac
Accent [†]	0.995 [0.96, 1.0]	22	[0.955, 1.0]	0.0
LOSO	0.677 (max)	20	—	0.0
Emo. (voice+token)	0.983 [0.93, 1.0]	19	[0.946, 1.0]	—
Emo. (told only)	0.995 [0.97, 1.0]	20	[0.964, 1.0]	0.0
Gender	0.992 [0.92, 1.0]	25	[0.912, 1.0]	0.0
Language	0.989 [0.97, 1.0]	19	[0.981, 0.997]	0.0
Emo. (voice)	0.617 [0.47, 0.77]	22	[0.477, 0.68]	0.35 100

[†]Speaker-confounded in-split; the LOSO row is the speaker-disjoint version — 67.7% is the layer max; the disciplined value is the L18–22 window mean, 61.0% (§4.4). Told-only’s acoustic baseline is the neutral-carrier control (below chance 0.20); voice+token shares the voice row’s input audio. The first revision’s single-split values (1.0 / 0.967 / 1.0 / 0.967 / 0.767) used smaller fixed test pools.

per emotion (emotional categories: 15 unique actors, 6M/14F clips, 13 strong / 7 normal intensity; neutral: 16 actors, 11M/9F); token mapping happy→elation, angry→anger, sad→sadness, surprise→surprise; max 160 tokens. Accent: VCTK mic1 via a Hugging Face parquet mirror; 3 speakers per accent (US p294/p297/p299; British p225/p226/p227; Indian p248/p251/p376; Irish p238/p245/p261, two of the three Northern Irish; speaker genders not recorded in our manifests); 20 clips per accent reading their own transcripts (max 180 tokens); split 12/6/2, with all three speakers shared across derive and test in every accent.

Robustness protocol (revision). Beyond the original single seed-0 split, every probe number is re-estimated as the mean over repeated stratified random splits (100 for language, 300 for the smaller axes) with 95% percentile bands, using the identical PCA(64)+ridge probe; each axis also gets a label-shuffled permutation null over matched splits (200 shuffled repeats for the 300-split axes). Accent adds a leave-one-speaker-out (LOSO) evaluation: train on two speakers per accent, test on the held-out third. Two capture additions: the tag route was re-captured with the carrier-audio bug fixed (the told-only condition: token on a speaker-matched neutral carrier; 20 clips per emotion), and gender was topped up from 14 to the full 20-clip derive pool per group ($n=40$). Vector geometry and language×gender orthogonality were recomputed at all 36 layers from raw one-vs-rest diff-of-means vectors.

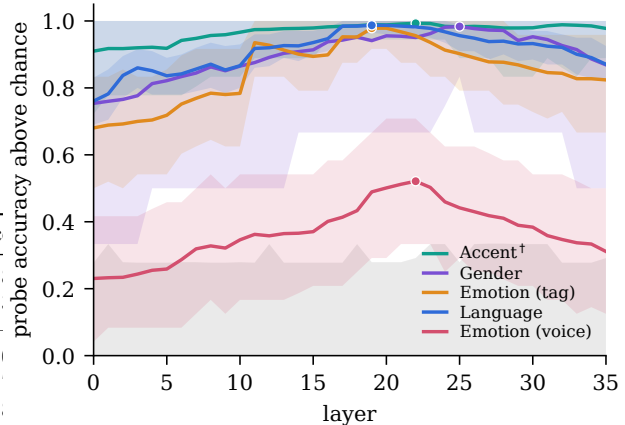


Figure 1. The identity ladder, cross-validated. Mean held-out probe accuracy over 100–300 repeated stratified splits, normalized above chance (chance: language 16.7%, gender 50%, emotion 20%, accent 25%); shaded bands are 95% percentile intervals, the gray floor is the label-shuffled permutation null (multi-class axes). Dots mark peak layers. [†]The accent curve remains speaker-confounded here; the speaker-disjoint version collapses to 67.7% (§4.4).

4. Results

4.1. The Identity Ladder

Figure 1 and Table 1 give the map. Reading it bottom-up:

Speaker-identity cues are available near the input. Gender is 87.7% readable at layer 0 (CV mean) and sits at or near ceiling from the mid-stack on (peak 99.2% [92–100]). The (confounded) accent probe likewise starts at 93.2% at layer 0. Whatever the network does with speaker identity, its linear traces are present early and consolidate mid-stack; gender’s Fisher separation peaks at layer 24, deeper than language’s (layer 19).

Content attributes assemble mid-stack. Language rises from 80.0% at layer 0 (CV mean) through a mid-stack plateau (peak 98.9% [97–100] at layer 19), and decays toward the output (89.2% at layer 35). Its Fisher separation peaks at layer 19 and collapses in the last two layers — the deepest layers appear to rotate toward output-token statistics rather than attribute structure (§4.3).

Felt emotion is the hardest listening task — at our n . Emotion inferred from an emotional reference recording starts at 38.4% (chance 20%) and averages only 61.7% [46.7–76.7] at its best layer (22) — our originally published 76.7% single split sat at the very top of that band, and the learning curve is still rising at

Table 2. The emotion 2×2 (peak CV probe accuracy; chance 20%). Rows: whether the model hears emotional audio; columns: whether it is told via its control token. Speaker-disjoint = repeated 3-actors-held-out CV (§4.5); the pattern survives it.

	Not told	Told ($\langle \text{emotion} \rangle$)
Neutral audio	(chance) 20%	99.5% [97–100]
Emotional audio	61.7% [47–77]	98.3% [93–100]
speaker-disjoint	64.8% [41–91]	100% / 98.9%

Speaker-disjoint row: voice-only 64.8% @L22; told-only 100% @L21; hear+tell 98.9% @L19 (means over 200 held-out-actor draws).

our maximum training size, so 61.7% is a lower bound rather than a ceiling (§4.5). The same emotions, specified through the model’s dedicated control token, are near-ceiling far earlier. That contrast is the subject of §4.2.

The ladder survives a corpus swap. A probe that only learned FLEURS recording artifacts would not survive new speakers, microphones, and speaking styles. It does: trained on all 600 FLEURS clips and tested on 870 in-the-wild Emilia clips of the same six languages (disjoint real speakers, podcast-style audio, captured under identical conditions in the prior sweep), the language probe reads 97.0% at layer 19 — the same peak layer, at nearly the within-corpus accuracy. Gender transfers even more sharply: the probe trained on 40 FLEURS clips reads the 80 VCTK clips (real speakers with official metadata genders) at 98.8%, and even the sign of projection onto the raw FLEURS diff-of-means direction classifies VCTK speakers at 84% — external validation that the dominant binary structure in §4.1 is gender, resolving a limitation the first revision could only flag (§5).

Figure 2 shows the same story geometrically: per-layer discriminant point clouds pull apart at view-specific depths (language slowly, over twenty layers; token-conditioned emotion almost immediately) and begin to re-merge near the output.

4.2. Two Routes, One (Coarse) Latent

An accidental control, kept and then fixed. The token condition was designed to place $\langle \text{emotion} \rangle$ on a neutral, speaker-matched carrier voice. Re-deriving the capture path while writing this paper revealed that the carrier’s audio was never loaded: the loop passed the emotional clip’s audio to both conditions, so the as-captured contrast is audio held constant, $\pm$ one token — accidentally a cleaner isolation of the token’s

Table 3. Independent-audio convergence at layer 6, calibrated (revision). Obs = $\cos(v_E^{\text{voice}}, v_E^{\text{told}})$; r_v/r_t = split-half reliability of each side (200 disjoint-half draws); Corr = attenuation-corrected obs/ $\sqrt{r_v r_t}$ ($\geq 1 \Rightarrow$ consistent with full alignment up to noise); $-g$ = the same cosine with each dataset’s gender direction projected out of both sides. Anisotropy-matched null at this layer: mean $|\cos| = 0.23$. Shared-audio peaks and the δ decomposition (angry $\cos(\delta, v_v) = +0.76$ at L6; sad -0.23) are in the text.

Emotion	Obs	r_v	r_t	Corr	$-g$
Angry	0.902	0.90	0.93	0.98	0.73
Disgust	0.878	0.70	0.90	1.11	0.68
Happy	0.863	0.79	0.91	1.02	0.69
Surprise	0.859	0.75	0.85	1.08	0.65
Fearful	0.765	0.85	0.51	1.16	0.64
Sad	0.440	0.36	0.17	1.81	0.34

marginal effect than the design. We kept that contrast under its true description, fixed the pipeline (with a regression test), re-captured the tag route as designed (the told-only condition), and gained what the accident had cost: a 2×2 over hearing and telling (Table 2) and a convergence measurement with fully independent audio.

The two conditions share their audio and differ by one prepended token (plus, for roughly half the clips, which of RAVDESS’s two fixed sentences serves as the reference transcript — a near-no-op), yet differ in every readability measure (Figure 3a): on the original split, peak probe accuracy 96.7% (layer 11) with the token vs. 76.7% (layer 22) without (CV: 98.3% vs. 61.7%), and peak Fisher separation 0.756 (layer 20) vs. 0.260 (layer 6) — roughly 3×. Each $\langle \text{emotion} \rangle$ tag is a single dedicated vocabulary token, an almost maximally clean conditioning signal; prepending it to a recording the model already hears still transforms how legibly the emotion is encoded. Hearing leaves affect smeared across the stack; being told snaps it into a linearly readable code.

Figure 3b asks where the token’s contribution goes. For each emotion we compare the direction the residual stream moves (relative to neutral) with and without the token; since the audio is shared, $v_E^{\text{tag}} = v_E^{\text{voice}} + \delta_E$ where δ_E is the token’s marginal displacement. The two directions align: angry at cosine 0.9755 at layer 6 ($\approx 12.7^\circ$ in a 2560-dim space), happy at 0.9273 (layer 5), surprise at 0.9205 (layer 6); the mean over all layers and emotions is 0.874 (the layer-6 mean is 0.905). And the contribution is large: the token-augmented mean-offset norm exceeds the voice-only one for every emotion at layer 18 (ratios 1.18–1.92 in the all-clip construction; the originally published 18-clip derive means

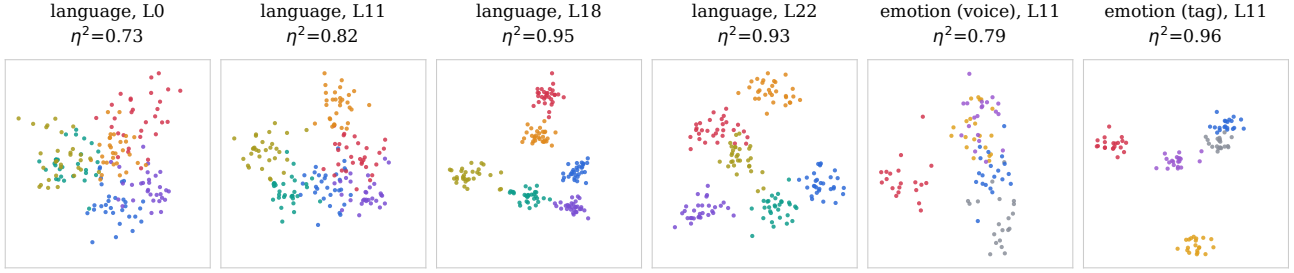


Figure 2. Watching separation happen. Per-clip mean residuals projected onto each layer’s top-2 between-class discriminant directions. Left four panels: language (180 clips, 6 classes) at increasing depth. Right two panels: the two emotion routes at layer 11 — the voice route (100 clips) is still smeared where the tag route (100 clips) has crystallized. η^2 is in-sample and shown for shape only.

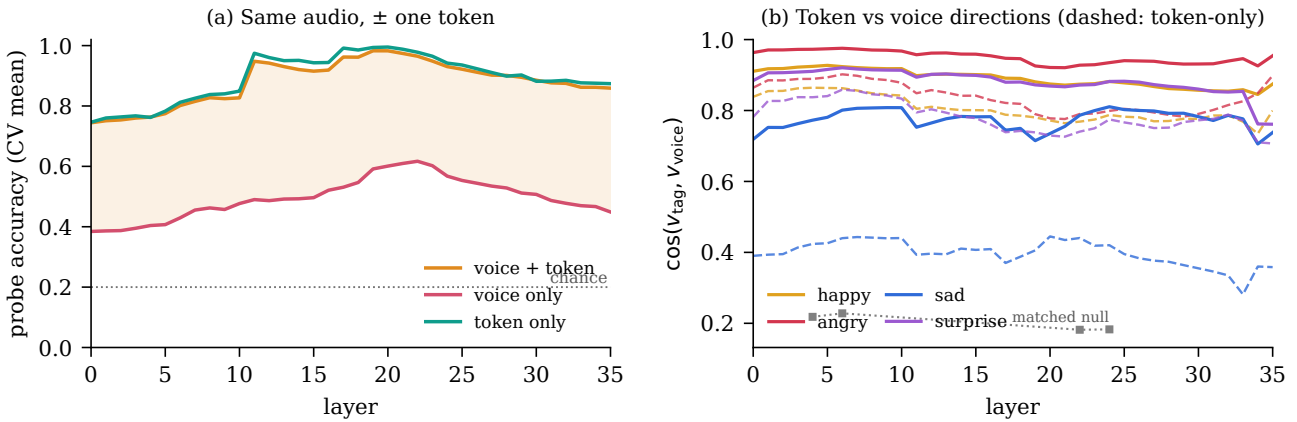


Figure 3. (a) CV-mean probe accuracy for the three emotion conditions (chance 0.2): voice + token (amber), voice only (rose), and — from the corrected-carrier capture — token only (teal), statistically indistinguishable from voice+token (§4.5). (b) Per-layer cosine between token and voice emotion directions. Solid: shared-audio construction (v^{tag} vs. v^{voice}). Dashed: independent audio (told-only vs. voice-only), unclipped; the dotted gray squares mark the anisotropy-matched null. Sad’s low dashed curve is a reliability artifact, resolved in Table 3 and Figure 4.

gave 1.38–2.55). The token does not open a second pathway; it reinforces, at roughly double strength, the very direction heard affect already writes.

Two qualifications. First, because $v_E^{\text{tag}} = v_E^{\text{voice}} + \delta_E$ by construction, part of the alignment is inherited; the interpretable signal is the contrast across emotions and layers, not the absolute level. Sad sits lowest here too (never above 0.811, with δ slightly anti-aligned, $\cos(\delta, v_{\text{voice}}) = -0.23$ at layer 6 vs. $+0.66$ to $+0.76$ for the others) — a pattern the reliability analysis below re-attributes to sad’s weak, noisy direction rather than to genuine disagreement. Second, both conditions use the same underlying RAVDESS material (a within-dataset contrast).

The told-only condition, and convergence without shared audio. The corrected re-capture gives the

told-only cell of Table 2. Two results. (i) Readability: told-only reads at 99.5% [96.7–100] at layer 20, 97.5% already by layer 11, and a full 100% under speaker-disjoint CV — statistically indistinguishable from voice+token (one discordant clip; §4.5). Emotional audio neither helps nor hurts a clean discrete instruction; the 2×2 ’s real structure is one bit, told vs. not told. (ii) Geometry: told-only directions share no audio with the voice-only directions, removing the $v^{\text{tag}} = v^{\text{voice}} + \delta$ construction entirely — and they still converge: angry 0.902 (layer 6), happy 0.864 (layer 4), surprise 0.859 (layer 6), Table 3. Two genuinely independent input routes — a discrete vocabulary token and heard prosody — write overlapping internal directions.

Calibrating the cosines — the convergence is real

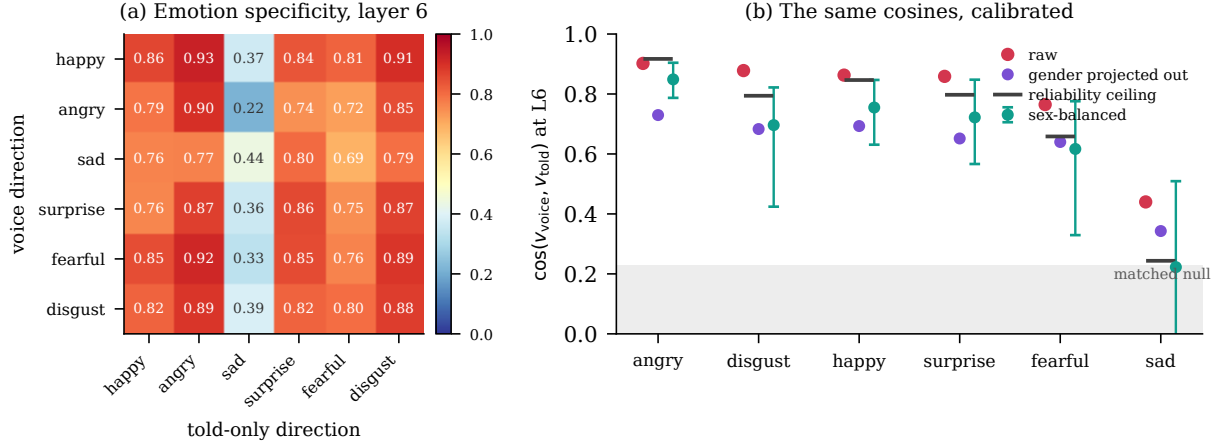


Figure 4. Calibrating the convergence. (a) The full voice $\times$ told cosine matrix at layer 6 (six emotions, independent audio). The diagonal exceeds the off-diagonal only modestly; the strongest off-diagonal entries (e.g. happy-voice $\cdot$ angry-told, 0.93) reveal a shared coarse expressive direction rather than per-emotion codes. (b) Per-emotion convergence at layer 6 under each control: raw, gender direction projected out, sex-balanced bootstrap (whiskers: 2.5–97.5%), with the split-half reliability ceiling ($\sqrt{r_v r_t}$, gray dashes) and the anisotropy-matched null (gray band). Sad’s raw cosine sits at its own reliability ceiling — the direction is weak, not misaligned.

but coarse. The revision subjects these cosines to the checks they previously lacked (Figure 4). First, the honest zero point: random same-size-subset diff-of-means pairs from these very datasets give mean $|\cos| \approx 0.23$ at layer 6 (anisotropy-matched null; the isotropic floor at $D=2560$ is 0.016), so raw cosines of 0.86–0.90 clear the null decisively — but the null is fourteen times higher than naive intuition. Second, specificity: the full voice $\times$ told matrix has a large off-diagonal — happy-voice aligns with angry-told at 0.926, above its own diagonal 0.863; the four-emotion diagonal mean beats the off-diagonal mean only modestly (0.77 vs. 0.68 at layer 6). Third, the sex-composition confound: the emotional draws are 6M/14F against neutral’s 11M/9F, and gender is the model’s single most readable direction; projecting each dataset’s own gender direction out of both sides (six-emotion analysis) drops angry from 0.902 to 0.73 (happy 0.86 $\rightarrow$ 0.69, surprise 0.86 $\rightarrow$ 0.65), and sex-balanced bootstrap resampling gives compatible values. The deepest version of the finding: the six heard-emotion directions are themselves mutually aligned at 0.72–0.96 at layer 6 — the layer-6 “emotion code” is essentially one emotional-expressiveness direction (relative to neutral) with small per-emotion tilts. The honest summary replaces “two mechanisms, one latent”: the token and the voice converge on a shared, coarse expressive direction — dominated by emotionality-vs-neutral, partly gender-shaped — while the per-emotion identity that probes read near-perfectly is a much smaller component on which the

two interfaces agree only partially (diagonal – off-diagonal gap ≈ 0.04 –0.08).

“Sad” was an artifact of unreliable directions, not a real exception. The earlier release headlined sad’s low independent-audio convergence (0.445) as a per-emotion dissociation. Split-half reliability dissolves it: sad’s own direction reproduces across disjoint clip halves at only 0.36 (voice) and 0.17 (told) at layer 6 — against 0.51–0.93 for the other emotions (all of whose corrected values land at ≈ 1) — and the attenuation-corrected convergence ($\text{obs}/\sqrt{r_a r_b}$) is consistent with full alignment up to noise (corrected ≥ 1 at every key layer). The same correction leaves the other emotions essentially at their raw values (Table 3). What is real about sad is that its direction is weak — the model moves little and inconsistently for sadness at $n=20$ — not that its two interfaces disagree. We retract the “revealing exception” framing.

Six emotions, same story — and a refuted prediction. A second capture extends both conditions with fearful and disgust (both single native tokens; same 15 RAVDESS actors). Both converge across routes (fearful raw 0.76, reliability-corrected 1.16; disgust raw 0.88, corrected 1.11 at layer 6), and the 6×6 voice $\times$ told matrix keeps the same signature — diagonal mean 0.78 vs. off-diagonal 0.74 at layer 6. We pre-registered (in the capture config) the prediction that a valence/arousal organization would put low-arousal disgust nearer sad; it does not — disgust patterns with the strong emotions, and the ordering across emotions

tracks direction reliability, not affect category. Convergence between hearing and telling is thus a robust, general property of this model’s emotion handling; per-emotion selectivity of the shared code is weak everywhere, and no affect category is a special case once reliability is accounted for.

4.3. Geometry: Non-Collinear Languages, Orthogonal Views

Languages own separate directions mid-stack. Measured at every layer: across layers 8–28 the worst of the 15 pairwise cosines among the six language one-vs-rest directions peaks at $|\cos| \approx 0.51$ (at layer 18 it is 0.446; e.g. en-zh -0.007 , ko-ja $+0.032$), with norms 35–51 at layer 18. This non-collinearity is a synthesis-conditioned property: comparable vectors derived from text-stream proxies in earlier work on the same model were near-collinear ($\cos \approx 0.99$).

The deepest layers explode. By layer 34 the same construction yields norms of 363–730 (≈ 13 – $19\times$ the layer-12 values) and the worst pairwise cosine rises to 0.66, with sign flips in several pairs (en-ko $-0.234 \rightarrow +0.158$; de-zh $\rightarrow +0.567$). Combined with the late-stack decay of probe accuracy and Fisher separation, we read layers 34–35 as reorganizing representations for token emission rather than maintaining attribute geometry. (The original release sampled geometry at layers 12/18/34; the revision computes it at all 36 layers.)

Language $\perp$ gender, at every layer. Across all 36 layers, the mean $|\cos|$ between the six language directions and the gender direction stays within $[0.113, 0.215]$ (raw diff-of-means vectors; minimum at layer 19). Projecting the gender direction out of the language directions preserves $\geq 97.0\%$ of their norm at every depth. (The originally published layer-20 value, 0.165, used τ -thresholded vectors; the raw-vector value there is 0.118, $\approx 83^\circ$.) To the extent voice cloning is compositional — any language in any voice — this near-orthogonality is the geometry one would expect to find, and does, everywhere in the stack.

All pairs, not one. The second revision extends the measurement to every attribute pair (full-stack one-vs-rest directions, all 36 layers; the full-stack language $\times$ gender range, $[0.128, 0.284]$, sits slightly above the derive-split estimate quoted above). Most pairs are comparably near-orthogonal — language $\times$ accent $[0.06, 0.15]$, accent $\times$ told-emotion $[0.04, 0.13]$, language $\times$ told-emotion $[0.08, 0.35]$ — with one instructive exception: gender $\times$ accent reaches $[0.16, 0.33]$, the least orthogonal pair measured. That is exactly where the corpus is confounded: our American accent group is all-female (§4.4), and the geome-

try sees it. Cross-corpus caveat: these directions come from different corpora and carriers, so the numbers measure subspace alignment, not within-distribution interaction.

4.4. The 100% We Declined to Headline — Measured

The accent probe reads 100% at every layer from 11 onward (in-split CV: 99.5% [96–100]). We flagged it as a diagnostic, not a finding: the corpus mirror provides only 3 speakers per accent, shared between train and test splits, so a probe can succeed by identifying who is speaking rather than how. The signature was in the shape: 93.2% already at layer 0 (CV), and the flattest Fisher profile of any view (range 0.156–0.221 — no consolidation depth).

The revision tests the confound directly with leave-one-speaker-out evaluation: train on two speakers per accent, test on the held-out third (three folds). Speaker-disjoint accuracy collapses to 67.7% at layer 20 (chance 25%; 41.7% at layer 0) — and, tellingly, the collapse restores the ladder’s shape: flat and modest through the early stack, then a genuine peak at layers 19–21, exactly where language consolidates. The second revision then disciplines this number too, since 67.7% is a max over 36 layers (winner’s curse): at the layer the in-split CV itself selects (22), LOSO reads 54.2%; the pre-specified L18–22 window mean is 61.0%; and a speaker-level permutation null (accent labels shuffled across the 12 speakers, max over layers 16–24 — a window containing it — 100 draws) puts its 97.5th percentile at 53.0%. The speaker-independent accent signal is real — the window mean clears the null, and the acoustic-feature baseline manages only 34.4% under the same LOSO (§4.5) — but “real and modest” means the window mean, not the layer-max. The honest accent story is therefore two findings: most of the 100% was speaker identity, and the speaker-independent remainder is real, modest, and mid-stack. Speaker-disjoint evaluation is the default protocol in paralinguistics (Schuller et al., 2009); our addition for the layer-resolved probing setting — where small- N convenience corpora make speaker leakage easy to miss — is the shape diagnostic: a probe that is flat-high from layer 0 is reading identity, and a collapse that restores depth structure marks the content that remains. A further caveat the revision surfaced: our American accent group is all-female (official VCTK metadata), so the in-split accent probe could also lean on gender; the LOSO number inherits this imbalance.

4.5. Validity: Baselines, Sensitivity, and What the Numbers Can Bear

Acoustic-feature baselines — is the ladder in the model or in the stimulus? A 39-dimensional model-free feature vector (13 MFCC means + stds, F0 statistics, spectral centroid/rolloff/flux, ZCR, log-energy) from each view’s actual input audio, probed with the identical CV protocol, separates stimulus from computation. Language: acoustics 79.7% vs. the model 98.9% — MFCCs carry much of language identity, the model adds the rest. Gender: 70.8% vs. 99.2%. Heard emotion: 34.6% (actor-disjoint 31.7%) vs. 61.7% — the model roughly doubles what shallow acoustics extract, so the 61.7% is not trivially “in the input”. Accent is the sharpest lesson: acoustics score 80.7% on speaker-overlapping splits but 34.4% under LOSO — shallow features carry almost no speaker-independent accent information, which makes the model’s 67.7% LOSO peak (against its own speaker-permutation null; below) a genuine model contribution. The told-only condition provides the designed control: its input audio is a neutral carrier, and acoustics predict the tag at 8.8% — at/below chance — while the model’s residuals read it at 99.5%. The token’s meaning is computed, not heard.

Real confidence intervals, and which ordering claims survive. The repeated-split bands report split-to-split variability, not clip-sampling uncertainty; pooled 5-fold CV (each clip tested exactly once) gives exact binomial CIs at the peak layers: language 99.2% [98.1, 99.7] ($n=600$), gender 100% [91.2, 100] ($n=40$), heard emotion 58.0% [47.7, 67.8] ($n=100$), voice+token 99.0% [94.6, 100], told-only 100% [96.4, 100], accent (in-split) 100% [95.5, 100]. Paired exact McNemar tests on the pooled predictions settle the orderings: told-only vs. voice+token differ on exactly one clip ($p=1.0$) — the earlier “most readable of all three” ranking is not statistically supported and we withdraw it; both token conditions beat voice-only on 41–42 discordant clips with zero reversals ($p<10^{-9}$). The 2×2 ’s real structure is one bit: told vs. not told.

Probe-choice sensitivity. Across PCA $k \in \{16, 32, 64, 128\}$ and ridge $\lambda \in \{0.1, 1, 10\}$ at each axis’s peak layer, λ is irrelevant (largest change from any λ swap: 6×10^{-4} ; PCA whitening) and k matters only at the low end ($k=16$ costs language 17 and heard emotion 10 points; $k \in \{64, 128\}$ agree within 0.2 points on every axis). A probe-free nearest-centroid classifier preserves every qualitative claim (told-only 89.3%, accent 81.3%, language 74.9%, heard emotion 47.8%): the ladder’s order is not an artifact of the PCA(64)+ridge choice.

Learning curves — ceiling or data starvation? Told-only saturates (92.3% at 4 training clips/class $\rightarrow$ 99.5% at 14). Heard emotion is still rising at our maximum (43.6% at 4 $\rightarrow$ 62.5% at 14, layer 22): 61.7% is a lower bound on heard-emotion readability, not a ceiling — a larger corpus would raise it, and claims about “the hardest listening task” are claims about our operating point.

5. Limitations

One model, one checkpoint, laptop-scale data. Cross-validation replaced all single-split claims and cost us one: voice-only emotion fell from 76.7% (single split) to a CV mean of 61.7% [46.7–76.7] — the original number was the top of its own band. Gender labels are taken from FLEURS metadata at face value, and gender’s permutation null is bimodal (97.5th percentile up to 0.917 across layers): with two tight clusters, any random binary labeling of the training set is either right or backwards, so within-corpus accuracy alone cannot distinguish “gender” from “some dominant binary structure”. The second revision resolves this externally: the FLEURS-trained probe (and even the raw direction’s projection sign) classifies VCTK speakers of known metadata gender at 98.8% / 84% (§4.1) — the dominant binary structure is gender. The η^2 values in Figure 2 are in-sample and inflated at small N ; we use them for shape only.

The carrier-audio discovery is narrated once, as method, in §4.2; what remains here is its residue: all voice+token results are stated for the as-captured audio-held-constant contrast, and the earlier project report’s “tag on a neutral voice” description of that run is superseded.

Further: the language residuals are reused from a monolingual-production capture of the same model rather than a fresh sweep; each condition is a single seeded capture run (generation is hardware-stochastic on MPS bf16, so re-capture reproduces distributions, not bytes); the emotion corpus is acted, North-American English, two fixed sentences — ecological validity for spontaneous or non-English affect is unknown; the emotional-clip draw is female-skewed (6 male / 14 female clips per emotion vs. the corpus’s balanced actors) — now measured as a confound and controlled in Table 3, not merely disclosed; and every measurement mean-pools the residual stream over the generated sequence, so token-position dynamics (when in generation an attribute becomes readable) are invisible to this instrument. Construction details worth knowing: the published shared-audio convergence curves (mac22) use derive-split means (18 clips)

while the revision’s δ analysis uses all 20 — the two agree within 0.046 everywhere; and the originally published orthogonality value used τ -thresholded vectors where the revision uses raw ones (0.165 vs. 0.118 at layer 20). And all results are correlational maps of what layers encode; causal use of these directions is a separate question, studied elsewhere with intervention methods and not claimed here.

6. Conclusion

Read passively, layer by layer, a voice-cloning audio LLM organizes the attributes of a voice on a ladder: identity-like cues (gender, speaker) are linearly present near the input and consolidate mid-stack; language assembles into a mid-network plateau before the output layers dissolve it; and emotion divides by interface — hearing alone averages 61.7% at its best layer, while the model’s own token, with or without any emotional audio underneath, is near-perfectly readable (98–99.5%, and 100% actor-disjoint). The two interfaces write overlapping directions even with fully independent audio, but the calibration this revision adds shows the shared latent is coarse — essentially one emotional-expressiveness direction with weak per-emotion selectivity — and that the one apparent per-emotion dissociation (sad) was an artifact of unreliable small- n directions. The views the model must recombine at will — language and gender — sit in near-orthogonal subspaces. The map is small- N and observational, but its structure (ordered depths, calibrated route convergence, orthogonal views, cross-corpus transfer, and two honestly-measured artifacts) offers a compact target for the causal question that naturally follows: does the model use what its layers so legibly encode?

Reproducibility

All numbers derive from the JSON outputs of the experiment folders (experiments/mac01/11/13/21/22/23/31; the first revision’s mac24 corrected-carrier capture, mac25 robustness suite, and mac26 gender top-up; and the second revision’s mac27 clip manifests, mac28 validity controls, mac29 external-validity suite, and mac32/33/34 emotion extension) of the higgs_tts_projection_v3 package. Figures are generated by paper_v2/make_figures.py from the same loader as the interactive post, which embeds the underlying per-clip data; experiments/rebuild_all.sh rebuilds analysis, figures, pages, and this PDF from the captured activations in order. Environments are pinned in environment/ (MPS venv lockfile +

creation script; analysis-env record). Every capture experiment records seeds, package versions, and generation parameters (temperature 0.7, top- p 0.95, seed 0) in its manifest; clip-level provenance (npz $\rightarrow$ source wav $\rightarrow$ speaker) is materialized by mac27. Analysis is deterministic given the archived activations and the pinned environment; capture itself is seeded but hardware-stochastic (MPS bf16 sampling), so re-capture reproduces distributions, not bytes. Capture used read-only forward hooks on all 36 layers, bf16, on a single Apple M1 Max.

Data and model licenses. FLEURS is CC-BY-4.0; RAVDESS is CC-BY-NC-SA-4.0 (download script with content verification in scripts/get_ravdess.sh); VCTK is ODC-By/CC-BY-4.0 via the mirror we pin (12 speakers, exact list in results/manifests/); the model is used under Boson AI’s research & non-commercial license (Boson AI, 2026).

Impact Statement

This is a read-only interpretability study of a publicly released voice-cloning model: we add no synthesis capability, release no cloned audio of real identities beyond corpus samples used under their licenses, and modify no weights. Voice cloning itself is dual-use — impersonation and fraud are the salient risks — and mapping where attribute information lives inside such models is, in our judgment, defense-relevant: the same maps that explain how a cloner represents speaker identity are the natural starting point for watermarking, attribution, and detection work. We do not publish steering recipes here; the intervention-based companion study is deliberately excluded.

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